



ECEN 314 - Signals and Systems

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Module 17 - Application: Analog Filter Design

Part 3 – More Chebyshev LPFs

+ transformations to HPFs and BPFs.

Learning Objectives



- Understand the form and characteristics of an Inverse Chebyshev low-pass magnitude response.
- Give specifications on the passband and the stop band, be able to design an Inverse Chebyshev LPF to meet those specifications.
- Be able to transform from a prototype LPF design to LPFs, HPFs, and BPFs.
- Given specifications on a BPF, be able to translate into equivalent specifications on a prototype LPF.

Chebyshev (Type II) LPFs



- If the ripple behavior in the passband of the Chebyshev filter is undesirable, there is a modified version of the Chebyshev filter that goes by the name of Chebyshev Type II or Inverse Chebyshev Filters.
- These filters have a smoother response in the passband and the ripple behavior has been moved to the stop-band.
- Chebyshev (Type I) LPF: $|H(f)|^2 = \frac{1}{1 + \varepsilon [C_n(f/f_1)]^2}$.
- Chebyshev (Type II) LPF: $|H(f)|^2 = \frac{1}{1 + \frac{1}{\varepsilon [C_n(f_2/f)]^2}}$.
- For the Type II LPF, the parameter ε is chosen to meet the passband/stopband edge constraints, then

$$|H(f_1)|^2 = \frac{1}{1 + \frac{1}{\varepsilon C_n^2(f_2/f_1)}} \geq \alpha_1 \rightarrow \varepsilon \geq \frac{1}{\left(\frac{1}{\alpha_1} - 1\right) C_n^2(f_2/f_1)}$$
$$|H(f_2)|^2 = \frac{1}{1 + \frac{1}{\varepsilon}} \leq \alpha_2 \rightarrow \varepsilon \leq \frac{1}{\frac{1}{\alpha_2} - 1}$$

Chebyshev (Type II) LPFs



- The lower bound must be smaller than the upper bound resulting in

$$\frac{1}{\left(\frac{1}{\alpha_1}-1\right)C_n^2(f_2/f_1)} \leq \frac{1}{\frac{1}{\alpha_2}-1} \rightarrow \left[C_n\left(\frac{f_2}{f_1}\right)\right]^2 \geq \frac{\frac{1}{\alpha_2}-1}{\left(\frac{1}{\alpha_1}-1\right)}.$$

- This leads to the same formula for the order of the Type I filter:

$$n \geq \cosh^{-1} \left(\sqrt{\left(\frac{1}{\alpha_2} - 1\right) / \left(\frac{1}{\alpha_1} - 1\right)} \right) / \cosh^{-1} \left(\frac{f_2}{f_1} \right).$$

- Once the order is selected, the parameter ε can be chosen to be within a range that will meet both the passband and stopband constraints

$$\frac{1}{\left(\frac{1}{\alpha_1} - 1\right) C_n^2\left(\frac{f_2}{f_1}\right)} \leq \varepsilon \leq \frac{1}{\frac{1}{\alpha_2} - 1}.$$

Chebyshev (Type II) LPFs



Example 17.1: Design an Inverse Chebyshev LPF to meet the following specifications:

Passband: $|f| < 1\text{kHz}$, $|H(f)|^2 > -1\text{dB}$,

Stopband: $|f| > 3\text{kHz}$, $|H(f)|^2 < -30\text{dB}$.

Chebyshev (Type II) LPFs



Example 17.1 (continued) : After some algebraic simplification, our designed magnitude response is:

$$|H(f)|^2 = \frac{9\left(\frac{f}{f_2}\right)^4 - 24\left(\frac{f}{f_2}\right)^2 + 16}{1500\left(\frac{f}{f_2}\right)^6 + 9\left(\frac{f}{f_2}\right)^4 - 24\left(\frac{f}{f_2}\right)^2 + 16}$$

The poles of this magnitude response are found to be at:

$$f = \pm j0.4965f_2, \text{ and } f = (\pm 0.4051 \pm j0.2095)f_2.$$

Converting to the s-plane the potential roots of the transfer function are at:

$$s = j2\pi f = \pm 3.1194f_2 \text{ and } s = (\pm 1.3164 \pm j2.5455)f_2$$

Choose the 3 in the LHP to be poles of the transfer function

$$\begin{aligned} p_1 &= -3.1194f_2 = -9358.3 \\ p_2 &= (-1.3164 + j2.5455)f_2 = -3949.1 + j7636.6 \\ p_3 &= p_2^* = -3949.1 - j7636.6 \end{aligned}$$

Chebyshev (Type II) LPFs



Example 17.1 (continued) :

$$|H(f)|^2 = \frac{9\left(\frac{f}{f_2}\right)^4 - 24\left(\frac{f}{f_2}\right)^2 + 16}{1500\left(\frac{f}{f_2}\right)^6 + 9\left(\frac{f}{f_2}\right)^4 - 24\left(\frac{f}{f_2}\right)^2 + 16}$$

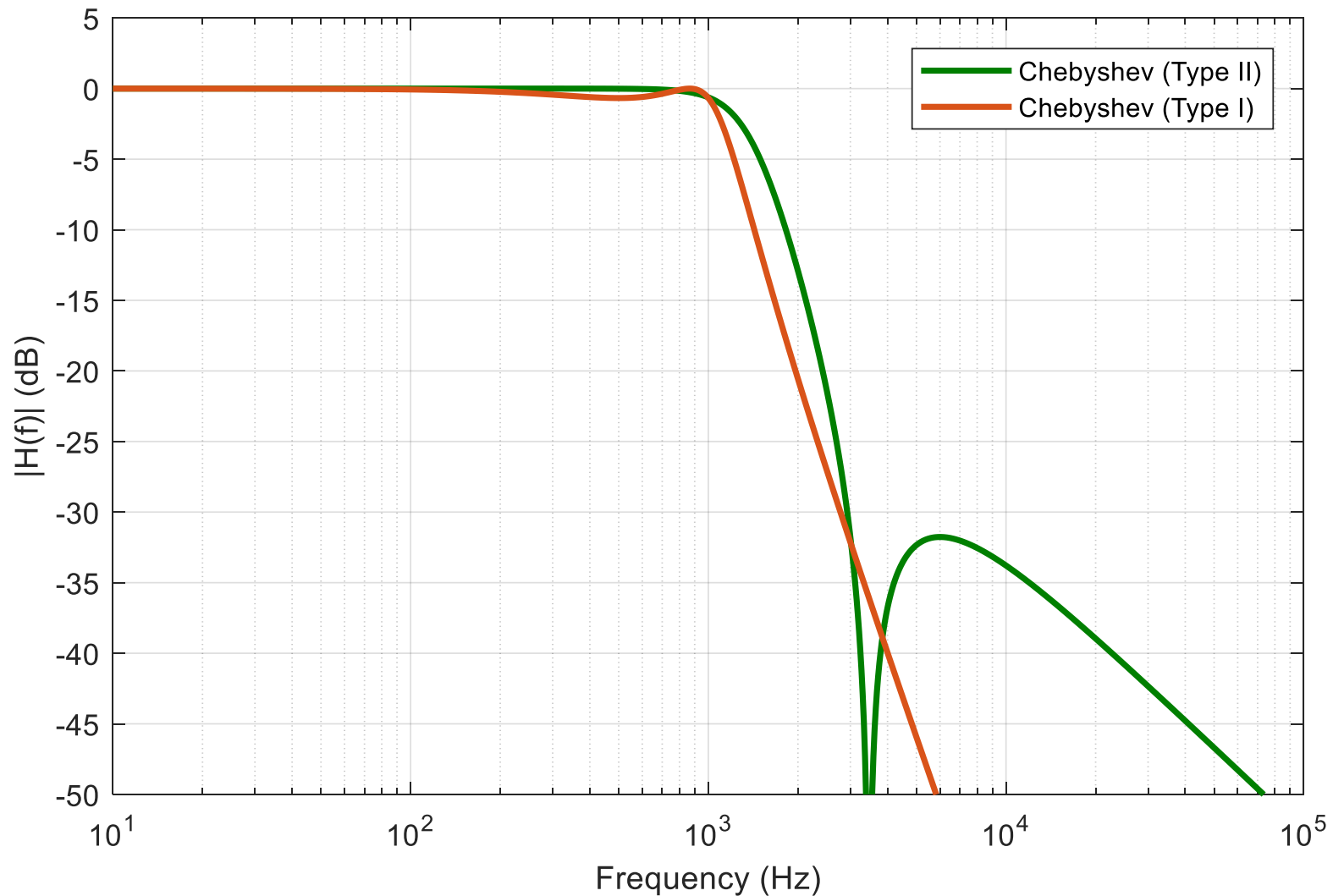
Note that our Type II LPF magnitude response also has four zeros which occur in the frequency domain at $f = \pm 1.1547f_2$ (both are double roots). In the s-plane the zeros are at $s = j2\pi f = \pm j7.2552f_2$

$$z_1 = j21,766, z_2 = -j21,766 = z_1^*$$

The transfer function can then be constructed as

$$H(s) = \frac{\left(1 - \frac{s}{z_1}\right)\left(1 - \frac{s}{z_1^*}\right)}{\left(1 - \frac{s}{p_1}\right)\left(1 - \frac{s}{p_2}\right)\left(1 - \frac{s}{p_2^*}\right)} = \frac{1 + \frac{s^2}{|21,766|^2}}{\left(1 + \frac{s}{9358.3}\right)\left(1 + \frac{s}{9358.2} + \frac{s^2}{|8597.3|^2}\right)}$$

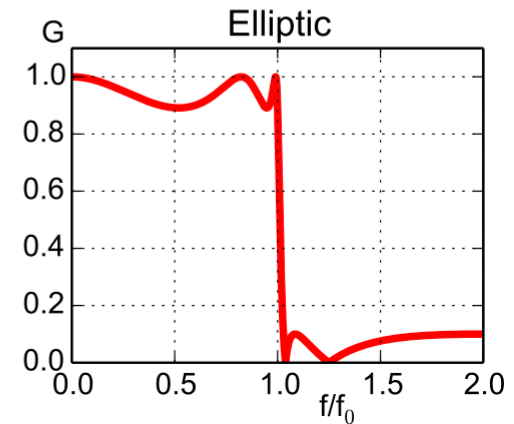
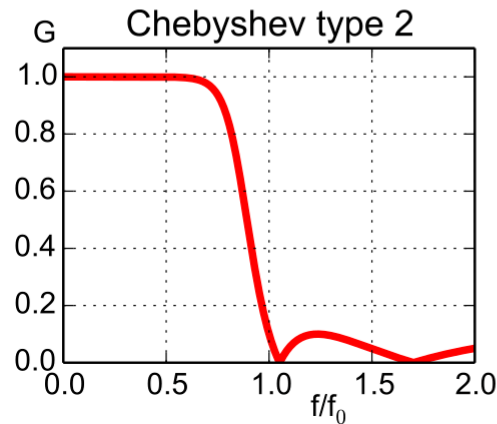
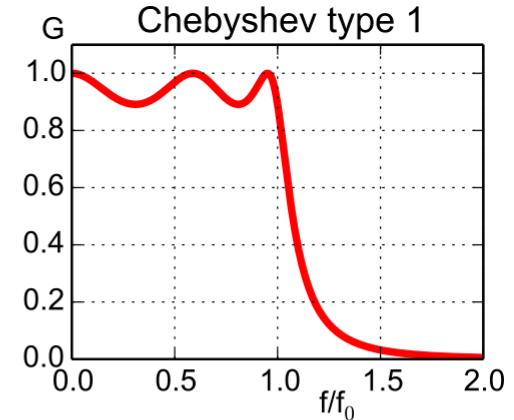
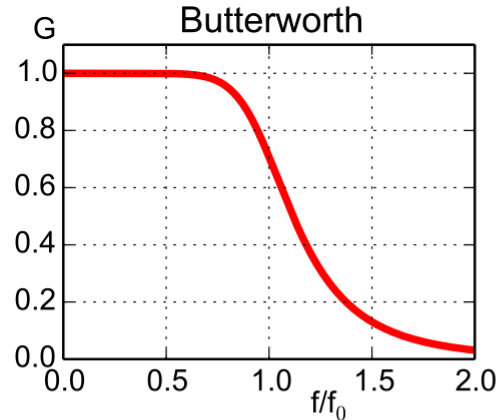
Chebyshev (Type II) LPFs



Elliptical Filters



- There is another class of filters known as elliptic filters which exhibit ripple behavior in both the passband and stop band but have a very quick transition between the two. The formulation of these filters is rather complicated so we won't go into it.
- These pictures show the generic behaviors of these various types of filters.



Analog Filter Transformations



- Once you know how to design a LPF, it is straightforward to transform a LPF to a HPF or a BPF.
- Let us suppose we have the design for a LPF in a normalized frequency domain (described by a normalized frequency variable u). For example, a Butterworth filter with a 3dB frequency of 1Hz would have a magnitude response

$$|H_P(u)|^2 = \frac{1}{1+u^{2n}}.$$

We refer to this as the prototype LPF design.

- This prototype design can be converted to various other forms as follows:

$$\text{LPF @ } f = f_o: \quad u \rightarrow \frac{f}{f_o} \quad |H_{LPF}(f)|^2 = \left| H_P \left(\frac{f}{f_o} \right) \right|^2$$

$$\text{HPF @ } f = f_o: \quad u \rightarrow \frac{f_o}{f} \quad |H_{HPF}(f)|^2 = \left| H_P \left(\frac{f_o}{f} \right) \right|^2$$

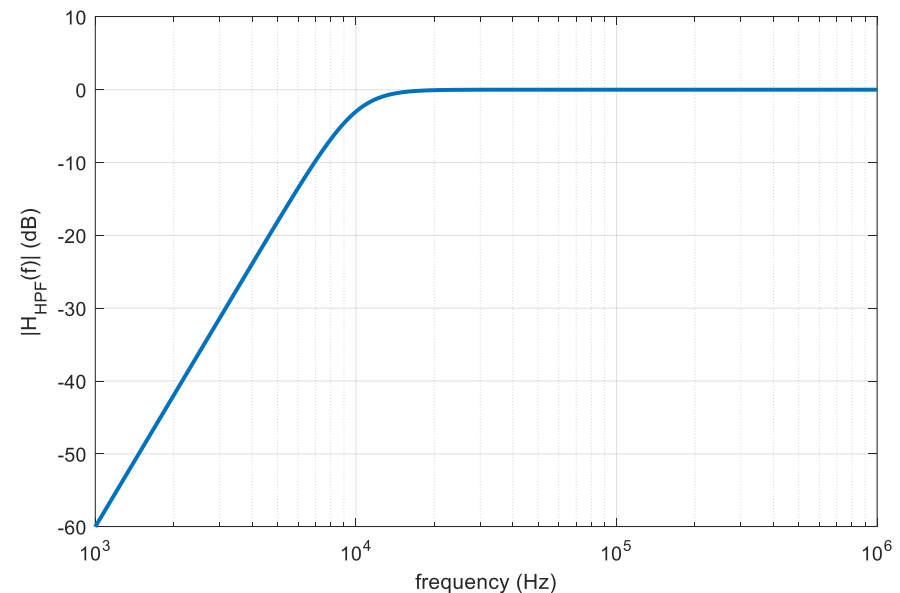
$$\text{BPF @ } f = f_o \text{ with BW}=B: \quad u \rightarrow \frac{f^2 - f_o^2}{fB} \quad |H_{BPF}(f)|^2 = \left| H_P \left(\frac{f^2 - f_o^2}{fB} \right) \right|^2$$

Analog Filter Transformations



- **Example 17.2:** Find the transfer function of a 3rd order Butterworth HPF with a 3dB frequency of 10kHz.

```
f=logspace(2,7,1000);  
fo=1e4;  
u=fo./f;  
H2=1./(1+u.^6);  
semilogx(f,10*log10(abs(H2)))  
axis([1e3 1e6 -60 10])  
grid on  
xlabel('frequency (Hz)')  
ylabel('|H_{HPF}(f)| (dB)')
```



Analog Filter Transformations



- **Example 7.3:** Find the transfer function of a 4th order Butterworth BPF with 3dB frequencies of 1kHz and 4kHz.

- 1) Start with a prototype 2nd order Butterworth LPF: $|H_P(u)|^2 = \frac{1}{1+u^4}$.

Note: the LPF to BPF transformation doubles the order of the filter.

- 2) Calculate the center frequency:

$$f_o = \sqrt{f_L f_H} = \sqrt{(1\text{kHz})(4\text{kHz})} = 2\text{kHz}.$$

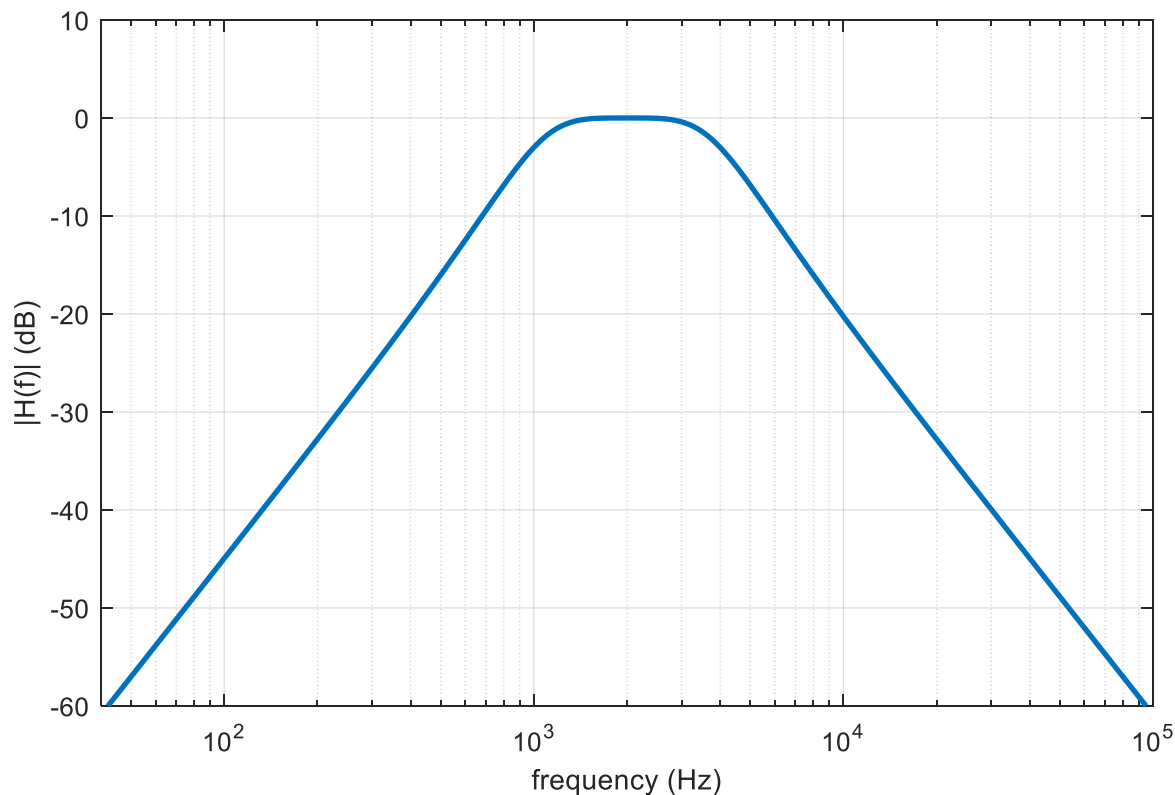
- 3) Calculate the BW:

$$B = (f_H - f_L) = (4\text{kHz} - 1\text{kHz}) = 3\text{kHz}.$$

- 4) Use the transformation $u \rightarrow \frac{f^2 - f_o^2}{fB}$

$$\begin{aligned} |H_{BPF}(f)|^2 &= \left| H_P \left(\frac{f^2 - f_o^2}{fB} \right) \right|^2 = \frac{1}{1 + \left(\frac{f^2 - f_o^2}{fB} \right)^4} = \frac{(fB)^4}{(fB)^4 + (f^2 - f_o^2)^4} \\ &= B^4 \frac{f^4}{f^8 - 4f_o^2 f^6 + (6f_o^4 + B^4)f^4 - 4f_o^6 f^2 + f_o^8} \end{aligned}$$

Analog Filter Transformations



```
>> fL=1e3;  
>> fH=4e3;  
>> fo=sqrt(fL*fH);  
>> B=fH-fL;  
>> f=logspace(1,6,1000);  
>> u=(f.^2-fo^2)./(f*B);  
>> H2=1./(1+u.^4);  
>> semilogx(f,10*log10(abs(H2)));  
>> axis([40 1e5 -60 10])  
>> grid on  
>> xlabel('frequency (Hz)')  
>> ylabel('|H(f)| (dB)')
```

Analog Filter Transformations



- **Example 17.3 (continued):** If it is desired to express the transfer function in the s-domain in terms of poles and zeros, proceed as follows:

- 5) Find the poles and zeros in terms of f .

$$|H(f)|^2 = \frac{3000^4 f^4}{f^8 - 4 \cdot 2000^2 f^6 + (6 \cdot 2000^4 + 3000^4) \cdot f^4 - 4 \cdot 2000^6 f^2 + 2000^8}$$

4 zeros @ $f = 0$

8 poles @ $f = \pm 3133.0 \pm 1603.5j$ and $f = \pm 1011.7 \pm 517.8j$

- 6) Translate to the s-plane using $s = j2\pi f$.

4 zeros @ $s = 0$

8 poles @ $s = \pm 10.075 \pm 19,685j$ and $s = \pm 3253 \pm 6357j$

- 7) Pick the poles in the LHP and half of the zeros to belong to $H(s)$.

2 zeros @ $s = 0$

4 poles @ $s = -10.075 \pm 19,685j$ and $s = -3253 \pm 6357j$

Analog Filter Transformations



➤ Example 17.3 (continued):

8) Construct $H(s)$ from the pole/zero locations:

$$H(s) = K \frac{s^2}{(s + 10,075 + j19,685)(s + 10,075 - j19,685)(s + 3,253 + j6,357)(s + 3,253 - j6,357)}$$

9) Choose the gain constant, K , to provide the correct gain in the passband. At the middle of the passband, the transfer function has a magnitude of

$$|H(j\omega_o)|^2 = \frac{K^2 \omega_o^4}{(10,075^2 + (\omega_o + 19,685)^2)(10,075^2 + (\omega_o - 19,685)^2)(3253^2 + (\omega_o + 6357)^2)(3253^2 + (\omega_o - 6357)^2)}$$

$$|H(j\omega_o)|^2 = K^2 \cdot 1.1684 \cdot 10^9$$

If we want the gain in the middle of the passband to be 1 (0dB), then choose K to be

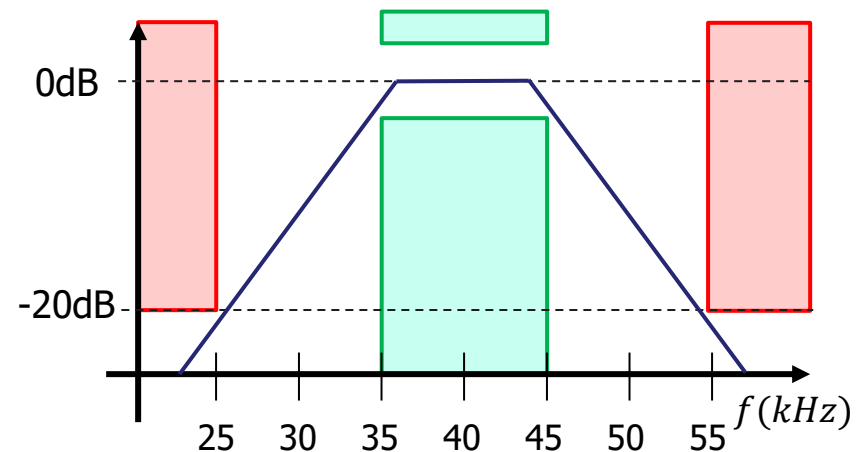
$$K = \frac{1}{\sqrt{1.1684 \cdot 10^9}} = 29,255.$$

$$\rightarrow H(s) = K \frac{(171.04s)^2}{(s^2 + 20,150s + 4.8902 \cdot 10^8)(s^2 + 6,506.9s + 5.0993 \cdot 10^7)}$$

Designing BPFs



- While choosing the parameters of a HPF (n and f_o for a Butterworth filter or n and ε for a Chebyshev filter) are very similar to the procedure for LPFs, finding the parameters for a BPF can be a bit more complicated.
- As an example, suppose we had to design a BPF to meet the following specifications:
 - ✓ Passband: $35\text{kHz} < f < 45\text{kHz}$, gain within $\pm 5\%$ of 0dB .
 - ✓ Stopbands: $f < 25\text{kHz}$ and $f > 55\text{kHz}$, gain $\leq -20\text{dB}$.



Step 1: Choose what kind of filter design to use. Since the math is easier, let's start with a Butterworth design.

Step 2: Find the transformation parameters.

$$\begin{aligned} f_L &= 35\text{kHz}, f_H = 45\text{kHz}, \\ \rightarrow f_o &= \sqrt{f_L f_H} = 39.69\text{kHz}, \\ B &= f_H - f_L = 10\text{kHz}. \end{aligned}$$

Designing BPFs



Step 3: Convert the various cutoff frequencies to the normalized prototype frequency scale. The LPF to BPF transformation is

$$u = \frac{f^2 - f_o^2}{fB}.$$

Right side:

$$\text{Passband edge: } f_1 = 45\text{kHz} \rightarrow u_1 = \frac{f_1^2 - f_o^2}{f_1 B} = 1.$$

$$\text{Stopband edge: } f_2 = 55\text{kHz} \rightarrow u_2 = \frac{f_2^2 - f_o^2}{f_2 B} = 2.6364.$$

Left side:

$$\text{Passband edge: } f_1 = 35\text{kHz} \rightarrow u_1 = \frac{f_1^2 - f_o^2}{f_1 B} = -1.$$

$$\text{Stopband edge: } f_2 = 25\text{kHz} \rightarrow u_2 = \frac{f_2^2 - f_o^2}{f_2 B} = -3.8.$$

Designing BPFs



Step 4: Design the prototype LPF

Consider designing a LPF to meet the constraints on the right edge:

$$u_1 = 1, u_2 = 2.64, \alpha_1 = 0.95^2 \text{ (-0.45dB)}, \alpha_2 = 0.01 \text{ (-20dB)}.$$

$$n = \frac{1}{2} \log \left(\frac{\frac{1}{\alpha_2} - 1}{\frac{1}{\alpha_1} - 1} \right) / \log \left(\frac{u_2}{u_1} \right) = \frac{1}{2} \log \left(\frac{99}{0.1080} \right) / \log(2.6364) = 3.52.$$

A 4th order filter is needed.

Now consider designing a LPF to meet the constraints on the left edge:

$$u_1 = -1, u_2 = -3.8, \alpha_1 = 0.95^2 \text{ (-0.45dB)}, \alpha_2 = 0.01 \text{ (-20dB)}.$$

$$n = \frac{1}{2} \log \left(\frac{\frac{1}{\alpha_2} - 1}{\frac{1}{\alpha_1} - 1} \right) / \log \left(\frac{u_2}{u_1} \right) = \frac{1}{2} \log \left(\frac{99}{0.1080} \right) / \log(3.8) = 2.55.$$

An 3rd order filter is needed.

We learned two things here:

- 1) The specifications on the right edge are stricter than the specifications on the left edge.
- 2) If we stick with the Butterworth design, we will need an 8th order BPF.

Designing BPFs



Perhaps we can get a lower order filter by using a Chebyshev design.

LPF to meet the constraints on the right edge:

$$u_1 = 1, u_2 = 2.64, \alpha_1 = 0.95^2 \text{ (-0.45dB)}, \alpha_2 = 0.01 \text{ (-20dB)}.$$

$$n = \cosh^{-1} \left(\sqrt{\left(\frac{1}{\alpha_2} - 1\right) / \left(\frac{1}{\alpha_1} - 1\right)} \right) / \cosh^{-1} \left(\frac{u_2}{u_1} \right) = 2.53$$

Now, only a 3rd order filter is needed.

HPF to meet the constraints on the right edge:

$$u_1 = -1, u_2 = -3.8, \alpha_1 = 0.95^2 \text{ (-0.45dB)}, \alpha_2 = 0.01 \text{ (-20dB)}.$$

$$n = \cosh^{-1} \left(\sqrt{\left(\frac{1}{\alpha_2} - 1\right) / \left(\frac{1}{\alpha_1} - 1\right)} \right) / \cosh^{-1} \left(\frac{u_2}{u_1} \right) = 2.04$$

Again, a 3rd order filter is needed.

Ultimately, our BPF based on a 3rd order Chebyshev LPF prototype will end up being a 6th order filter.

Alternatively, we can build an 8th order BPF based on a 4th order Butterworth LPF prototype.

Designing BPFs



In order to choose the value of the parameter ε in our Chebyshev design, focus attention on the right edge constraints. Designing a 3rd order Chebyshev LPF would proceed as follows:

$$|H_{BPF}(f)|^2 = \frac{1}{1 + \varepsilon [C_3(u)]^2} \bigg|_{u = \frac{f^2 - f_o^2}{fB}}$$

$$\text{Passband edge: } f = f_1: |H_P(u_1)|^2 = \frac{1}{1 + \varepsilon} \geq 0.95^2 \rightarrow \varepsilon \leq 0.108.$$

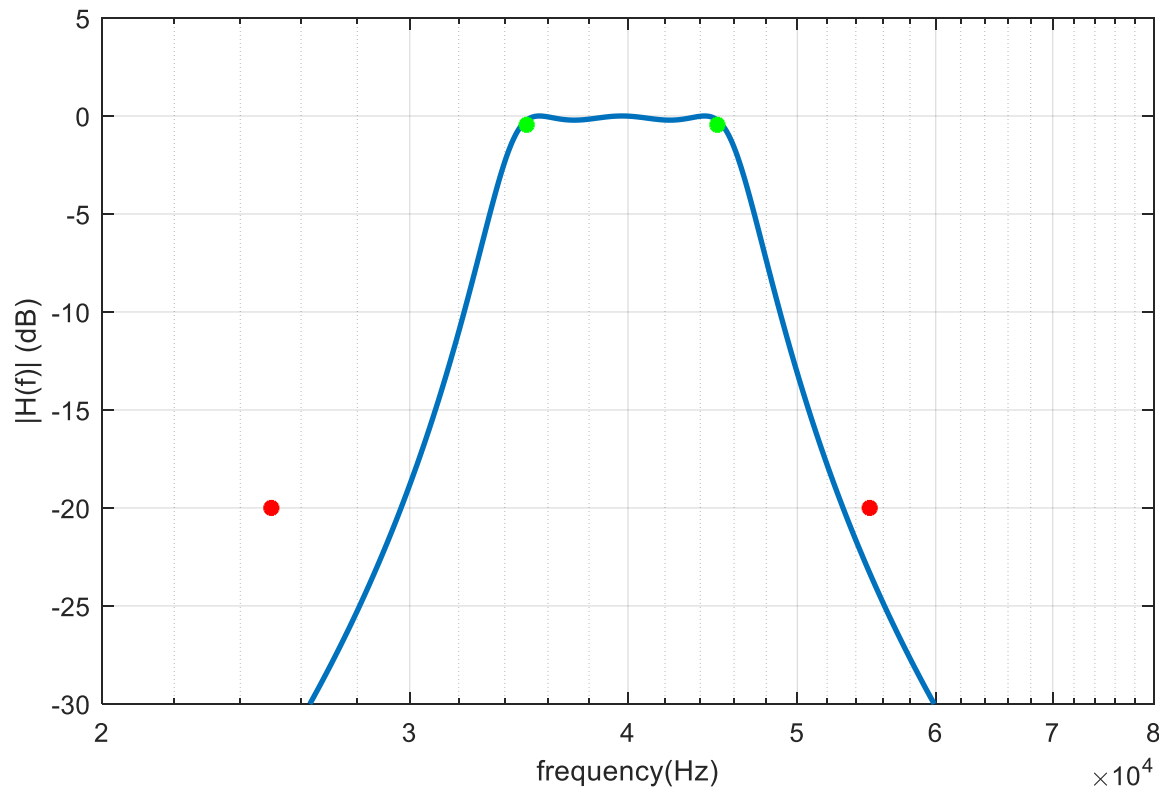
$$\text{Stopband edge: } f = f_2: |H_P(u_2)|^2 = \frac{1}{1 + \varepsilon [C_3(u_2)]^2} \leq 0.01 \rightarrow \varepsilon \geq 0.0232.$$

Picking an intermediate value, let's choose $\varepsilon = 0.05 = \frac{1}{20}$.

Step 5: Perform the transformation on the prototype LPF to form the BPF.

$$|H_P(u)|^2 = \frac{1}{1 + \frac{1}{20} [C_3(u)]^2}$$
$$\rightarrow |H_{BPF}(f)|^2 = \left| H_P \left(\frac{f^2 - f_o^2}{fB} \right) \right|^2 = \frac{1}{1 + \frac{1}{20} \left[C_3 \left(\frac{f^2 - 1.5753 \cdot 10^9}{f 10^4} \right) \right]^2}$$

Designing BPFs



```
>> fL=35e3; fH=45e3;  
>> fo=sqrt(fL*fH);  
>> B=fH-fL;  
>> f=logspace(4,5,1000);  
>> u=(f.^2-fo^2)./(f*B);  
>> C3=4*u.^3-3*u;  
>> H2=1./(1+C3.^2/20);  
>> semilogx(f,10*log10(abs(H2)))  
>> axis([20e3 80e3 -30 5])  
>> grid on  
>> xlabel('frequency(Hz)')  
>> ylabel('|H(f)| (dB)')
```